

Quiz #06

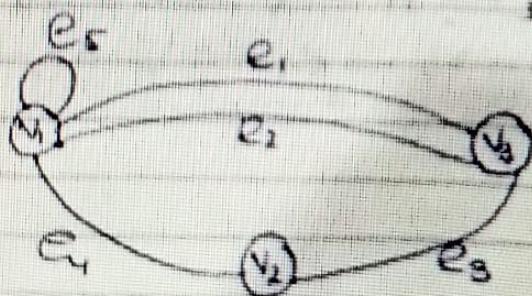
Section CSJ

Date

Section-J

$$A = \begin{matrix} & v_1 & v_2 & v_3 \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \end{matrix} & \begin{bmatrix} 1 & 1 & 2 \\ 1 & 0 & 1 \\ 2 & 1 & 0 \end{bmatrix} \end{matrix}$$

(a) Make a graph of this



(b)

$$\begin{bmatrix} 1 & 1 & 2 \\ 1 & 0 & 1 \\ 2 & 1 & 0 \end{bmatrix} \times \begin{bmatrix} 1 & 1 & 2 \\ 1 & 0 & 1 \\ 2 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1+1+4 & 1+0+2 & 2+1+0 \\ 1+0+2 & 1+0+1 & 2+0+0 \\ 2+1+0 & 2+0+0 & 4+1+0 \end{bmatrix}$$

$$= \begin{bmatrix} 6 & 3 & 3 \\ 3 & 2 & 2 \\ 3 & 2 & 5 \end{bmatrix} \times \begin{bmatrix} 1 & 1 & 2 \\ 1 & 0 & 1 \\ 2 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 6+3+6 & 6+0+3 & 12+3+0 \\ 3+2+3 & 3+0+2 & 6+2+0 \\ 3+2+10 & 3+0+5 & 6+2+0 \end{bmatrix}$$

$$= \begin{bmatrix} 15 & 9 & 15 \\ 8 & 5 & 8 \\ 15 & 8 & 8 \end{bmatrix}$$

(c) $v_3 \rightarrow e_1 \rightarrow v_1 \rightarrow e_5 \rightarrow v_1 \rightarrow e_4 \rightarrow v_2$

$$v_3 \rightarrow e_2 \rightarrow v_1 \rightarrow e_5 \rightarrow v_1 \rightarrow e_4 \rightarrow v_2$$

$$v_3 \rightarrow e_1 \rightarrow v_1 \rightarrow e_1 \rightarrow v_3 \rightarrow e_3 \rightarrow v_2$$

$$v_3 \rightarrow e_2 \rightarrow v_1 \rightarrow e_2 \rightarrow v_3 \rightarrow e_3 \rightarrow v_2$$

$$v_3 \rightarrow e_1 \rightarrow v_1 \rightarrow e_2 \rightarrow v_3 \rightarrow e_3 \rightarrow v_2$$

$$v_3 \rightarrow e_2 \rightarrow v_1 \rightarrow e_1 \rightarrow v_3 \rightarrow e_3 \rightarrow v_2$$

$$v_3 \rightarrow e_3 \rightarrow v_2 \rightarrow e_4 \rightarrow v_1 \rightarrow e_4 \rightarrow v_2$$

$$v_3 \rightarrow e_3 \rightarrow v_2 \rightarrow e_3 \rightarrow v_3 \rightarrow e_3 \rightarrow v_2$$